WEEK 3 · LECTURE L2 · UNIT I

Business Intelligence Fundamentals

What a warehouse, a metric and a dashboard actually are, and why so many organizations have no single source of truth.

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AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

- Explain the layers of a BI stack: sources, pipeline, warehouse, metric layer, dashboard.
- Explain why a metric is an agreement, and specify the six questions that define one.
- Apply Ackoff, Inmon and Kimball, Goodhart and Campbell, and Wang and Strong to HR data.
- Evaluate why a single source of truth is a political as much as a technical achievement.

COURSE OUTCOMES

CLO 1 (BI fundamentals) and CLO 4 (metrics and KPIs).

THE BI STACK

Five layers between a transaction and a decision

Layer	What it does	Common failure in HR
Sources	Systems where work happens: HRIS, payroll, ATS, LMS	Each optimized for today's transaction, not history
Pipeline	Extracts, cleans, conforms and loads data	Data re-keyed by hand into spreadsheets
Warehouse	Keeps history in one place, organized for questions	Absent: last March is overwritten
Metric layer	Agreed definitions with owners	Each team defines its own
Dashboard	Shows decisions, not data	Bought before the layers beneath it exist

EXAM LENS

A dashboard bought before the warehouse and metric layer exist will display numbers nobody trusts.

A WAREHOUSE IS A COPY, KEPT ON PURPOSE

Sources, pipeline, warehouse

Operational systems

Payroll, the HRIS, the ATS: built to run a process today, not to remember last year.

Extract, conform, load

Force data onto shared definitions and keys. The conforming step is where the politics lives.

History that does not move

A separate store, organized for asking questions, that remembers what was true in March after March is overwritten.

A METRIC IS AN AGREEMENT, NOT A FORMULA

Six questions that turn "attrition rate" into a definition

Question	Choices to settle
Numerator	All leavers or voluntary only? Retirements? End of fixed term?
Denominator	Opening, closing or average headcount? Contractors in or out?
Period	Financial year or rolling twelve months? Annualized from a quarter?
Grain	Whole firm, department, site or manager? At what size do you suppress?
Owner	Who may change the definition, and who must be told?
Restatement	When it changes, is history recalculated or left as published?

EXAM LENS

Two teams using different answers will report different numbers and both will be "right". Name the six questions in any answer on HR metrics.

A DASHBOARD IS A DECISION SURFACE

What makes a dashboard useful

Every tile needs an owner and an action

If nobody is accountable and nothing follows from the number moving, the tile is decoration.

Comparison beats level

A bare number cannot be interpreted; meaning comes from comparison with last period, a target or a peer unit.

A watched number gets managed

Publish time to fill, and time to fill will improve, possibly at the cost of quality of hire (Goodhart).

THE SINGLE SOURCE OF TRUTH

Why so many organizations have none

What it means

Every report draws on one agreed record for each employee: ask two managers for headcount and get the same number.

Two systems of record

An old on-premise HRIS and a new cloud HRIS both holding employee records produce conflicting truths.

The spreadsheet paradox

Universal use of private spreadsheets with low satisfaction is not success; it is the fallback when no system is trusted.

EXAM LENS

Agreeing one definition of an employee means somebody's version of the truth must lose. That makes it a political achievement.

DATA QUALITY

Real HR data is never clean

Impossible records

A course marked "not started" that still carries an assessment score.

Records that disagree

Age and tenure that cannot both be true for the same person.

Labels that do not match

System names in one table that match nothing in another.

Write down every cleaning rule you apply. A figure nobody can reproduce is an opinion.

EXERCISE

Define time to hire honestly

Start the clock

When the vacancy opens, when it is approved, or when it is advertised?

Stop the clock

Offer made, offer accepted, or first day at work?

Split it

By role, location and source channel; one average can hide a slow channel.

Choose the summary

Mean or median? One extreme vacancy can move a mean but not a median.

THEORY IN DEPTH · ACKOFF (1989)

From data to wisdom

THE CORE IDEA

Data are symbols; information answers who, what, where; knowledge answers how; wisdom judges what is worth doing.

HOW IT WORKS

Each level adds context and purpose. A dashboard sits at information; a decision needs knowledge.

LIMITS AND CRITIQUE

The neat pyramid is criticized: in practice the levels blur, and judgment shapes even raw data.

IN PRACTICE

An organization can hold vast data in spreadsheets but little information, because nobody has agreed what the numbers mean.

EXAM LENS

Use the hierarchy to explain why more data does not automatically mean better decisions, then note the critique.

THEORY IN DEPTH · INMON (1992); KIMBALL (1996)

Two ways to build a warehouse

THE CORE IDEA

Inmon: build one integrated enterprise warehouse first, then departmental marts. Kimball: build business-focused marts with shared dimensions.

HOW IT WORKS

Kimball's star schema joins a fact table of events to dimension tables (who, when, where).

LIMITS AND CRITIQUE

Inmon is slower but consistent; Kimball is faster but needs discipline on shared definitions.

IN PRACTICE

An HR fact table of appraisal and attrition events joined to an employee dimension table is a Kimball design.

EXAM LENS

Compare the two approaches on speed, consistency and governance; neither is simply better.

THEORY IN DEPTH · GOODHART (1975); CAMPBELL (1976)

A measure that becomes a target stops measuring

THE CORE IDEA

When people are rewarded on a number, they change their behavior to move the number.

HOW IT WORKS

Campbell: the more a quantitative indicator drives social decisions, the more it is corrupted and corrupts the process it monitors.

LIMITS AND CRITIQUE

Not every target fails; pairing a metric with a counter-metric limits gaming.

IN PRACTICE

Reward recruiters on time to hire and quality of hire may fall; track both together.

EXAM LENS

Pair Goodhart with a counter-metric in any answer that proposes HR KPIs.

THEORY IN DEPTH · WANG AND STRONG (1996)

Data quality has four families

THE CORE IDEA

Quality is fitness for use, judged by the people who use the data.

HOW IT WORKS

Intrinsic (accuracy), contextual (relevance, timeliness), representational (clarity), accessibility (can it be reached).

LIMITS AND CRITIQUE

The dimensions overlap, and fitness depends on the task, so no single score fits all uses.

IN PRACTICE

Contradictory learning records fail intrinsic quality; inconsistent system labels fail representational quality.

EXAM LENS

Classify each data problem into a family; this shows analytical rigour rather than a list of complaints.

PRACTICE FOR THE END-TERM EXAMINATION

Questions to rehearse

"A metric is an agreement, not a formula." Explain this statement and discuss its implications for building HR dashboards. **20 marks**

A strong answer: covers the six defining questions; links to the metric layer and single source of truth; applies Goodhart and Campbell; proposes counter-metrics.

Compare the Inmon and Kimball approaches to data warehousing and evaluate which suits an HR function beginning its analytics journey. **20 marks**

A strong answer: explains both clearly; compares on speed, consistency and governance; reaches a justified, conditional recommendation.

Explain Wang and Strong's four families of data quality with an HR example of each. **10 marks**

A strong answer: names all four families with a correct, distinct example for each.

CARRY THESE INTO THE EXAMINATION

Three things to remember

1. **A warehouse keeps history on purpose**
   That is its power and its risk.

2. **A metric is an agreement with an owner**
   Not a formula in a cell.

3. **A dashboard should change a decision**
   If it changes nothing, it is decoration.

SET READINGS · FROM THE LESSON PLAN

Read for this lecture

- Kavanagh, M. J., & Johnson, R. D. (2018), Ch 2 and Ch 14.

COMPANION

Each set reading is explained in the matching Unit Reader on rksingh.tech/materials.

THEORY SOURCES · FOR REFERENCE

Sources for this lecture

Works named on the slides of this lecture. They are background sources, not set readings.

Ackoff, R. L. (1989). From data to wisdom. *Journal of Applied Systems Analysis, 16*, 3–9.

Campbell, D. T. (1976). *Assessing the impact of planned social change* (Occasional Paper Series No. 8). Public Affairs Center, Dartmouth College.

Goodhart, C. A. E. (1975). Problems of monetary management: The U.K. experience. In *Papers in monetary economics* (Vol. 1). Reserve Bank of Australia.

Inmon, W. H. (1992). *Building the data warehouse*. QED Technical Publishing Group.

Kimball, R. (1996). *The data warehouse toolkit*. Wiley.

Wang, R. Y., & Strong, D. M. (1996). Beyond accuracy: What data quality means to data consumers. *Journal of Management Information Systems, 12*(4), 5–33. https://doi.org/10.1080/07421222.1996.11518099